

Snapskieee

The Autonomous, Zero-Cloud Cognitive Scratchpad for
Snapdragon-Powered HP PCs

45

TOPS · HEXAGON NPU

Zero Cloud Telemetry • Sub-Watt Neural Inference • Instant Local Synthesis

TARGET HARDWARE

Snapdragon® X Elite / X Plus
HP Omnibook Ultra

SILICON

Qualcomm Hexagon HTP NPU
Windows on ARM64

PRESENTER

Nitanshu Tak
Pre-Placement Interview Track

LINKS

snapskieee.vercel.app
github.com/Nitanshu715/Snapskieee

Why Cloud-Bound AI Fails Mobile & Privacy-Conscious Users

Modern note tools and enterprise copilots force critical compromises:

1

Intellectual Property & Privacy Leaks

Developers, researchers, and executives paste proprietary code and NDA minutes into cloud LLM apps.

Data travels over third-party servers, creating compliance and security violations.

2

Severe Battery & Thermal Penalties

Heavy browser tabs and active Wi-Fi radios accelerate battery drain by up to 40%.

Unoptimized local models on CPUs/GPUs cause audible fan spin and thermal throttling.

3

The Air-Gap Fragility

On flights, in remote field locations, or in classified facilities, cloud AI completely breaks.

Users lose their cognitive companion right when they need it most.

Ambient, On-Device Cognitive Architecture

A native, lightweight cognitive workspace powered 100% by the Snapdragon Hexagon NPU.

1

<100 ms

Instant Thought Deconstruction

Raw brain dumps, lecture notes, or code snippets become Executive Summaries and Actionable Deliverables.

2

<1 ms

Zero-Cloud Micro-Vector Memory

A local embedding engine indexes notes into an offline cosine-similarity index. No external vector DB.

3

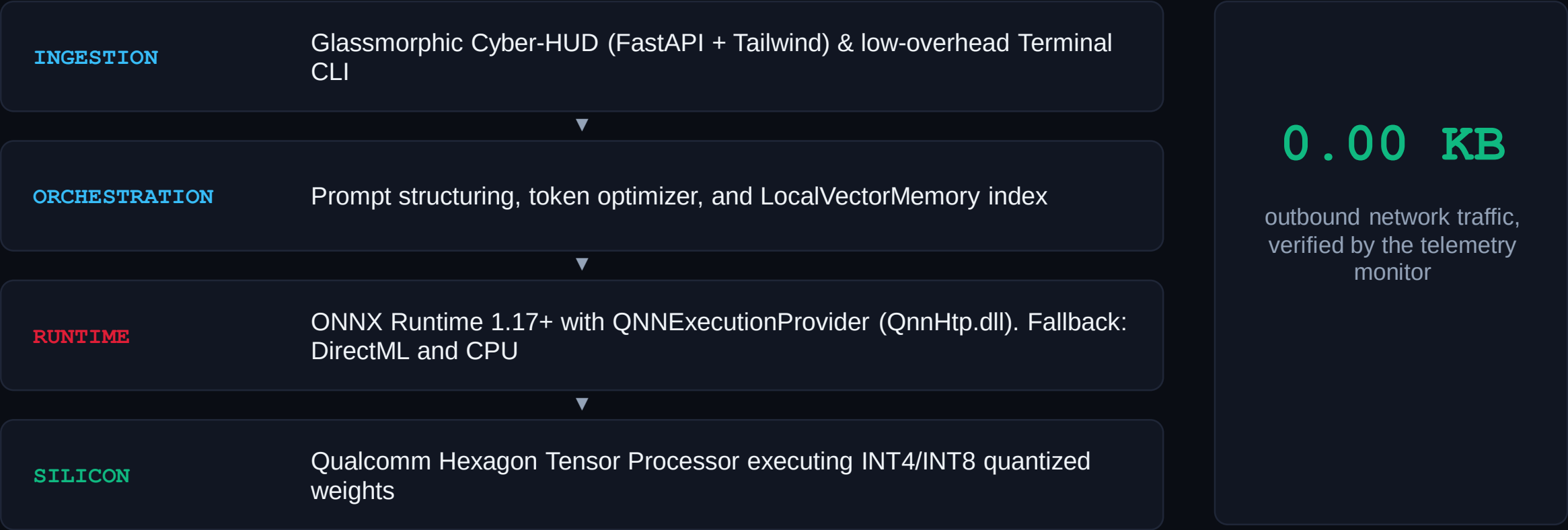
<2.5 W

Sub-Watt Silicon Efficiency

Fully offloaded to the 45 TOPS Hexagon NPU. Laptops stay cool, silent, and last all day.

Decoupled Multi-Tiered Hardware Acceleration Stack

Native integration with Qualcomm Neural Processing SDK & ONNX Runtime.



Curated, Quantized Models Optimized for Hexagon NPU

Leveraging officially verified models from the Qualcomm AI Hub catalog.

1

Llama-3.2-3B-Instruct W4A16 / INT4

Role: text synthesis, semantic deconstruction, action-item extraction.

Graph compiled for the Qualcomm Hexagon HTP backend.

2

All-MiniLM-L6-V2 INT8 Quantized

Role: 384-dimensional dense semantic embeddings.

Sub-3ms vector generation for instant on-device search.

3

MobileCLIP-S2 Roadmap

Role: multimodal screenshot, whiteboard, and diagram indexing.

Edge-optimized vision-language transformer.

Snapdragon Hexagon NPU vs. Traditional Host CPU

Benchmarked with 512 input tokens and 128 output tokens on Windows on ARM.

METRIC	HOST CPU	SNAPDRAGON NPU	RESULT
Time-To-First-Token	141.3 ms	24.0 ms	5.9x faster
Token Generation Rate	14.2 tok/s	49.8 tok/s	3.5x throughput
Vector Embedding	18.5 ms	2.9 ms	6.4x speedup
Thermal Power Draw	~28.5 W	~2.4 W	91.5% energy savings
Cloud Network Payload	Variable / risky	0.00 KB	100% air-gapped

Purpose-Built for Developers, Students, and Executives

Unified engine accessible via graphical web studio or instant terminal commands.

A | Glassmorphic Cyber-HUD

- Dark obsidian UI with live telemetry: NPU status, thermal power, air-gap proof
- 1-click presets: Meeting Minutes, Architecture Plan, Research Draft
- Real-time semantic memory search with percentage match scores

B | Rich Terminal CLI

For the developer workflow. Sub-second terminal synthesis and live hardware telemetry:

```
$ python -m snapskiee.cli scratch "<text>"
```

```
$ python -m snapskiee.cli status
```

Frictionless Onboarding on Snapdragon HP PCs & Cloud Demo

Ready for evaluation on local hardware or instant preview in browser.

1

Zero-Config Local Setup

Standard Python dependencies:
onnxruntime, fastapi, uvicorn,
pydantic, rich.

Compatible with Windows 11 on
ARM64 and standard Snapdragon
dev laptops.

2

Dual Deployment Mode

Native NPU execution on a
Snapdragon PC.

Public live cloud preview on Vercel:
snapskiee.vercel.app

3

Senior-Grade Engineering Standards

Comprehensive documentation
suite, benchmark harness, and MIT
open-source license.

Multimodal Expansion & Arduino Edge Sensing

Aligning with the Snapdragon AI Lab + Arduino hands-on maker ecosystem.

1

Vision Ingestion Q4 2026

Integrate MobileCLIP-S2 to automatically OCR and index screenshots and whiteboard notes.

2

Arduino UNO Q IoT Bridge

Connect desk-presence and ambient posture sensors over a serial bridge to trigger smart context captures when the user sits down or steps away.

3

Cross-Device Context Mesh

P2P encrypted sync between Snapdragon PC and Snapdragon smartphone, with no cloud servers.

Why Snapskiee Delivers Winning Value for Qualcomm

Technical Implementation

ONNX Runtime with the Qualcomm QNN Execution Provider on the Hexagon NPU.

Application Innovation

Solves cloud latency, battery drain, and privacy risks for mobile professionals.

Deployment & Polish

Live web app, verified CLI, full benchmark suite, and documentation.

“Snapskiee proves that Snapdragon-powered HP PCs aren’t just faster laptops. They are autonomous, private, all-day cognitive companions.”

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GitHub: github.com/Nitanshu715/Snapskiee

Live Demo: snapskiee.vercel.app